

Applications of ferrous and non-ferrous alloys

Module IV

FERROUS ALLOYS

Ferrous alloys—those in which iron is the prime constituent—are produced in larger quantities than any other metal type.

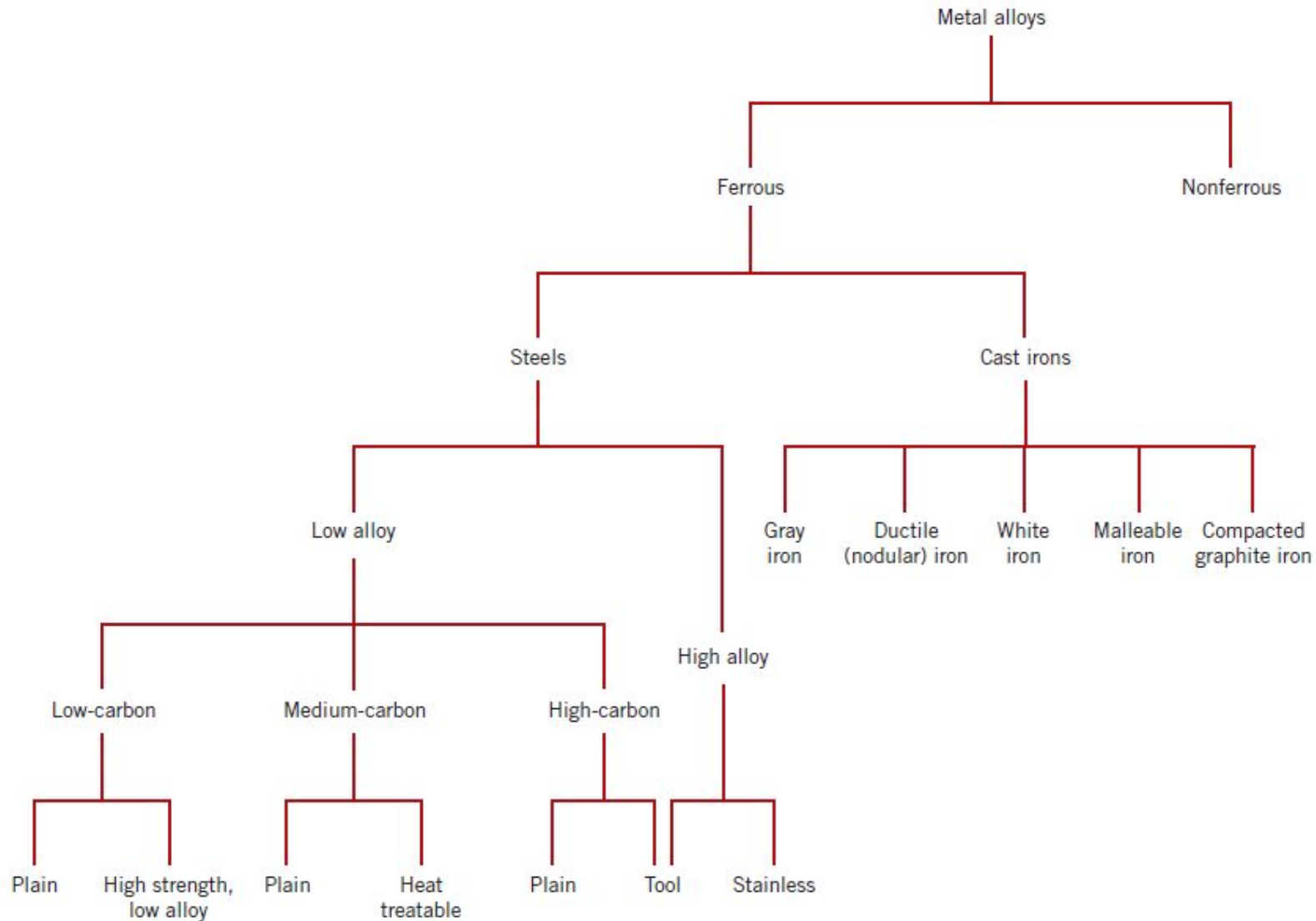


Figure 11.1 Classification scheme for the various ferrous alloys.

STEELS

- Steels are iron–carbon alloys with varying concentrations of other alloying elements.
- There are thousands of steel alloys with different compositions and heat treatments.
- Mechanical properties are influenced by the carbon content, typically less than 1.0 wt%.
- Common steels are classified into low-, medium-, and high-carbon types based on carbon concentration.
- Subclasses within each group exist, differentiated by the concentration of other alloying elements.
- Plain carbon steels have residual impurities and a small amount of manganese.
- Alloy steels intentionally contain additional alloying elements in specific concentrations.

LOW-CARBON STEELS

Plain-low carbon steels

- Most widely produced steels fall within the low-carbon classification.
- Low-carbon steels typically have less than about 0.25 wt% carbon.
- They are unresponsive to heat treatments for martensite formation, relying on cold work for strengthening.
- Microstructures of low-carbon steels consist of ferrite and pearlite constituents.
- These alloys are relatively soft and weak but exhibit outstanding ductility and toughness.
- They are machinable, weldable, and the least expensive to produce among all steels.
- Common applications include automobile body components, structural shapes (I-beams, channel, and angle iron), and sheets used in pipelines, buildings, bridges, and tin cans.
- They typically have a yield strength of 275 MPa tensile strengths between 415 and 550 MPa, and a ductility of 25%EL.

LOW-CARBON STEELS ...

High-strength, low-alloy (HSLA) steels.

- Another category of low-carbon alloys is the high-strength, low-alloy (HSLA) steels.
- HSLA steels contain alloying elements like copper, vanadium, nickel, and molybdenum.
- Alloy concentrations can be as high as 10 wt%, resulting in higher strengths compared to plain low-carbon steels.
- Most HSLA steels can be strengthened through heat treatment, achieving tensile strengths exceeding 480 MPa (70,000 psi).
- These alloys are ductile, formable, and machinable.

LOW-CARBON STEELS -COMPOSITION

<i>Designation^a</i>		<i>Composition (wt%)^b</i>		
<i>AISI/SAE or ASTM Number</i>	<i>UNS Number</i>	<i>C</i>	<i>Mn</i>	<i>Other</i>
<i>Plain Low-Carbon Steels</i>				
1010	G10100	0.10	0.45	
1020	G10200	0.20	0.45	
A36	K02600	0.29	1.00	0.20 Cu (min)
A516 Grade 70	K02700	0.31	1.00	0.25 Si
<i>High-Strength, Low-Alloy Steels</i>				
A572 Grade 42	—	0.21	1.35	0.30 Si, 0.20 Cu (min)
A633 Grade E	K12002	0.22	1.35	0.30 Si, 0.08 V, 0.02 N, 0.03 Nb
A656 Type 3	—	0.18	1.65	0.60 Si, 0.08 V, 0.02 N, 0.10 Nb

^aThe codes used by the American Iron and Steel Institute (AISI), the Society of Automotive Engineers (SAE), and the American Society for Testing and Materials (ASTM), and in the Uniform Numbering System (UNS) are explained in the text.

^bAlso a maximum of 0.04 wt% P, 0.05 wt% S, and 0.30 wt% Si (unless indicated otherwise).

LOW-CARBON STEELS -STRENGTH

<i>AISI/SAE or ASTM Number</i>	<i>Tensile Strength [MPa (ksi)]</i>	<i>Yield Strength [MPa (ksi)]</i>	<i>Ductility [%EL in 50 mm (2 in.)]</i>	<i>Typical Applications</i>
<i>Plain Low-Carbon Steels</i>				
1010	325 (47)	180 (26)	28	Automobile panels, nails, and wire
1020	380 (55)	210 (30)	25	Pipe; structural and sheet steel
A36	400 (58)	220 (32)	23	Structural (bridges and buildings)
A516 Grade 70	485 (70)	260 (38)	21	Low-temperature pressure vessels
<i>High-Strength, Low-Alloy Steels</i>				
A572 Grade 42	415 (60)	290 (42)	24	Bolted/rivet bridges & buildings
A633 Grade E	515 (75)	380 (55)	23	Structures used at low ambient temperatures
A656 Type 3	655 (95)	552 (80)	15	Truck frames & railway cars

MEDIUM-CARBON STEELS

- Medium-carbon steels have carbon concentrations ranging from about 0.25 to 0.60 wt%.
- Heat treatment involves austenitizing, quenching, and tempering to enhance mechanical properties.
- Microstructures of tempered martensite are typical in the tempered condition.
- Plain medium-carbon steels have low hardenabilities, limiting successful heat treatment to thin sections with rapid quenching rates.
- Chromium, nickel, and molybdenum additions improve heat-treatability, resulting in various strength–ductility combinations.
- Heat-treated medium-carbon alloys are stronger than low-carbon steels but may sacrifice ductility and toughness.
- Applications include railway wheels and tracks, gears, crankshafts, machine parts, and high-strength structural components requiring a balance of high strength, wear resistance, and toughness.

MEDIUM-CARBON STEELS COMPOSITION

AISI/SAE Designation^a	UNS Designation	Composition Ranges (wt% of Alloying Elements in Addition to C)^b			
		Ni	Cr	Mo	Other
10xx, Plain carbon	G10xx0				
11xx, Free machining	G11xx0				0.08–0.33 S
12xx, Free machining	G12xx0				0.10–0.35 S, 0.04–0.12 P
13xx	G13xx0				1.60–1.90 Mn
40xx	G40xx0			0.20–0.30	
41xx	G41xx0		0.80–1.10	0.15–0.25	
43xx	G43xx0	1.65–2.00	0.40–0.90	0.20–0.30	
46xx	G46xx0	0.70–2.00		0.15–0.30	
48xx	G48xx0	3.25–3.75		0.20–0.30	
51xx	G51xx0		0.70–1.10		
61xx	G61xx0		0.50–1.10		0.10–0.15 V
86xx	G86xx0	0.40–0.70	0.40–0.60	0.15–0.25	
92xx	G92xx0				1.80–2.20 Si

^aThe carbon concentration, in weight percent times 100, is inserted in the place of “xx” for each specific steel.

^bExcept for 13xx alloys, manganese concentration is less than 1.00 wt%.

Except for 12xx alloys, phosphorus concentration is less than 0.35 wt%.

Except for 11xx and 12xx alloys, sulfur concentration is less than 0.04 wt%.

Except for 92xx alloys, silicon concentration varies between 0.15 and 0.35 wt%.

AISI/SAE/ASTM SPECIFICATION OF STEELS

- Alloyed medium-carbon steels and their compositions are detailed in the table on the previous slide.
- Designation schemes for these steels involve the Society of Automotive Engineers (SAE), the American Iron and Steel Institute (AISI), and the American Society for Testing and Materials (ASTM).
- The AISI/SAE designation is a four-digit number: the first two digits indicate alloy content, and the last two denote carbon concentration.
- Plain carbon steels have initial digits 1 and 0, while alloy steels use other two-digit combinations (e.g., 13, 41, 43).
- The third and fourth digits represent the weight per cent carbon multiplied by 100.
- For instance, a 1060 steel is a plain carbon steel containing 0.60 wt% C.

HIGH-CARBON STEELS...

- High-carbon steels typically have carbon contents ranging between 0.60 and 1.4 wt%.
- They are the hardest, strongest, and least ductile among carbon steels.
- High-carbon steels are commonly used in a hardened and tempered condition, offering exceptional wear resistance and a sharp cutting edge.
- Tool and die steels, falling into the high-carbon category, often contain chromium, vanadium, tungsten, and molybdenum.
- Alloying elements combine with carbon to form hard and wear-resistant carbide compounds (e.g., Cr_23C_6 , V_4C_3 , WC).
- The following table lists some tool steel compositions and their applications.
- Tool steels find use in cutting tools, dies for forming and shaping materials, knives, razors, hacksaw blades, springs, and high-strength wire.

HIGH-CARBON STEELS

Table 11.2b Typical Applications and Mechanical Property Ranges for Oil-Quenched and Tempered Plain Carbon and Alloy Steels

<i>AISI Number</i>	<i>UNS Number</i>	<i>Tensile Strength [MPa (ksi)]</i>	<i>Yield Strength [MPa (ksi)]</i>	<i>Ductility [%EL in 50 mm (2 in.)]</i>	<i>Typical Applications</i>
<i>Plain Carbon Steels</i>					
1040	G10400	605–780 (88–113)	430–585 (62–85)	33–19	Crankshafts, bolts
1080 ^a	G10800	800–1310 (116–190)	480–980 (70–142)	24–13	Chisels, hammers
1095 ^a	G10950	760–1280 (110–186)	510–830 (74–120)	26–10	Knives, hacksaw blades
<i>Alloy Steels</i>					
4130	G41300	830–1550 (120–225)	670–1340 (97–195)	24–11	Gears, bolts, axles
4340	G43400	980–1960 (142–284)	895–1570 (130–228)	21–11	Bushings, aircraft tubing
6150	G61500	815–2170 (118–315)	745–1860 (108–270)	22–7	Shafts, pistons, gears

^aClassified as high-carbon steels.

HIGH-CARBON STEELS: TOOL STEELS

Table 11.3 Designations, Compositions, and Applications for Six Tool Steels

<i>AISI Number</i>	<i>UNS Number</i>	<i>Composition (wt%)^a</i>						<i>Typical Applications</i>
		<i>C</i>	<i>Cr</i>	<i>Ni</i>	<i>Mo</i>	<i>W</i>	<i>V</i>	
M1	T11301	0.85	3.75	0.30 max	8.70	1.75	1.20	Drills, saws; lathe and planer tools
A2	T30102	1.00	5.15	0.30 max	1.15	—	0.35	Punches, embossing dies
D2	T30402	1.50	12	0.30 max	0.95	—	1.10 max	Cutlery, drawing dies
O1	T31501	0.95	0.50	0.30 max	—	0.50	0.30 max	Shear blades, cutting tools
S1	T41901	0.50	1.40	0.30 max	0.50 max	2.25	0.25	Pipe cutters, concrete drills
W1	T72301	1.10	0.15 max	0.20 max	0.10 max	0.15 max	0.10 max	Blacksmith tools, woodworking tools

^aThe balance of the composition is iron. Manganese concentrations range between 0.10 and 1.4 wt%, depending on alloy; silicon concentrations between 0.20 and 1.2 wt%, depending on the alloy.

Source: Adapted from *ASM Handbook*, Vol. 1, *Properties and Selection: Irons, Steels, and High-Performance Alloys*, 1990. Reprinted by permission of ASM International, Materials Park, OH.

STAINLESS STEELS

- Stainless steels are highly resistant to corrosion, particularly in diverse environments like the ambient atmosphere. Chromium is the main alloying element, requiring a concentration of at least 11 wt% Cr. Nickel and molybdenum additions can enhance corrosion resistance.
- Stainless steels are categorized into three classes based on the predominant microstructure phase: **martensitic, ferritic, or austenitic**. Table (next slide) provides details on various stainless steels.
- Martensitic stainless steels can be heat-treated to have martensite as the primary microconstituent. Alloying elements in significant concentrations alter the iron–iron carbide phase diagram.
- Austenitic stainless steels extend the austenite phase field to room temperature and are not heat-treatable. Austenitic stainless steels are the most corrosion-resistant due to high chromium and nickel contents.
- Ferritic stainless steels are composed of the α -ferrite (BCC) phase and are also not heat-treatable.
- Both martensitic and ferritic stainless steels are magnetic; austenitic stainless steels are not.
- Some stainless steels are used at elevated temperatures and in severe environments, resisting oxidation and maintaining mechanical integrity. Applications include gas turbines, high-temperature steam boilers, heat-treating furnaces, aircraft, missiles, and nuclear power–generating units.
- Specific austenitic steels, like 316L, are used in biomedical applications for temporary orthopaedic devices.
- The following Table also includes ultra-high-strength stainless steel (17-4PH) with exceptional

FERRITIC AND
AUSTENITIC
STAINLESS STEELS

Table 11.4 Designations, Compositions, Mechanical Properties, and Typical Applications for Austenitic, Ferritic, Martensitic, and Precipitation-Hardenable Stainless Steels							
AISI Number	UNS Number	Composition (wt%) ^a	Condition ^b	Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Ferritic							
409	S40900	0.08 C, 11.0 Cr, 1.0 Mn, 0.50 Ni, 0.75 Ti	Annealed	380 (55)	205 (30)	20	Automotive exhaust components, tanks for agricultural sprays
446	S44600	0.20 C, 25 Cr, 1.5 Mn	Annealed	515 (75)	275 (40)	20	Valves (high temperature), glass molds, combustion chambers
Austenitic							
304	S30400	0.08 C, 19 Cr, 9 Ni, 2.0 Mn	Annealed	515 (75)	205 (30)	40	Chemical and food processing equipment, cryogenic vessels
316L	S31603	0.03 C, 17 Cr, 12 Ni, 2.5 Mo, 2.0 Mn	Annealed	485 (70)	170 (25)	40	Welding construction, temporary biomedical orthopedic devices

MARTENSITIC AND PRECIPITATION HARDENABLE STAINLESS STEELS

<i>Martensitic</i>							
410	S41000	0.15 C, 12.5 Cr, 1.0 Mn	Annealed Q & T	485 (70) 825 (120)	275 (40) 620 (90)	20 12	Rifle barrels, cutlery, jet engine parts
440A	S44002	0.70 C, 17 Cr, 0.75 Mo, 1.0 Mn	Annealed Q & T	725 (105) 1790 (260)	415 (60) 1650 (240)	20 5	Cutlery, bearings, surgical tools
<i>Precipitation Hardenable</i>							
17-4PH	S17400	0.07 C, 16.25 Cr, 4 Ni, 4 Cu, 0.3 (Nb + Ta), 1.0 Mn, 1.0 Si	Precipitation hardened	1310 (190)	1172 (170)	10	Chemical, petrochemical, and food- processing equipment, aerospace parts

^aThe balance of the composition is iron.

^bQ & T denotes quenched and tempered.

DUPLEX STEELS...

1. Introduction to Duplex Steels:

- I. **Definition:** Duplex steels are a type of stainless steel with a two-phase microstructure consisting of both austenitic (face-centred cubic) and ferritic (body-centred cubic) phases.
- II. **Composition:** Typically composed of 22-26% chromium, 4.5-6.5% nickel, 3-4% molybdenum, and 0.1-0.3% nitrogen.

2. Microstructure:

- I. **Dual Phases:** Austenitic and ferritic phases coexist, providing a combination of properties from both structures.
- II. **Benefits:** Enhanced strength, corrosion resistance, and toughness compared to single-phase stainless steels.

3. Key Properties:

I. Corrosion Resistance:

- i. Resistant to general and localized corrosion.
- ii. Suitable for aggressive environments like marine and chemical processing.

II. Mechanical Properties:

- i. Higher strength than austenitic and ferritic stainless steels.
- ii. Good toughness and ductility.

III. Weldability:

- i. Generally good weldability, but attention to heat input is important.
- ii. Precipitation of harmful phases may occur if not properly controlled.

DUPLEX STEELS

4. Application:

I. Oil and Gas Industry:

- i. Used in offshore platforms for corrosion resistance and strength.
- ii. Pipelines and components exposed to aggressive environments.

II. Chemical Processing:

- i. Chemical reactors and vessels handling corrosive substances.
- ii. Pumps, valves, and fittings.

III. Desalination Plants:

- i. Corrosion-resistant material for seawater applications.
- ii. Used in heat exchangers and piping systems.

5. Challenges and Considerations:

I. Phase Instability:

- i. Care is required during fabrication and welding to avoid detrimental phase transformations.

II. Cost Considerations:

- i. Generally more expensive than conventional stainless steel.
- ii. Economic viability depends on the specific application.

CAST IRONS...

1. Generic Definition:

- Cast irons are a class of ferrous alloys with carbon contents above 2.14 wt%.
- Most cast irons contain between 3.0 and 4.5 wt% C and other alloying elements.

2. Melting and Casting:

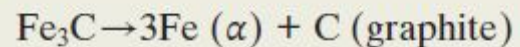
- Alloys within the composition range of 3.0-4.5 wt% C become completely liquid at temperatures between 1150°C and 1300°C.
- Lower melting temperatures than steels make cast irons easily melted and suitable for casting.

3. Brittleness and Fabrication:

- Some cast irons are very brittle.
- Casting is the most convenient fabrication technique due to the brittleness of certain cast-iron types.

4. Cementite Decomposition:

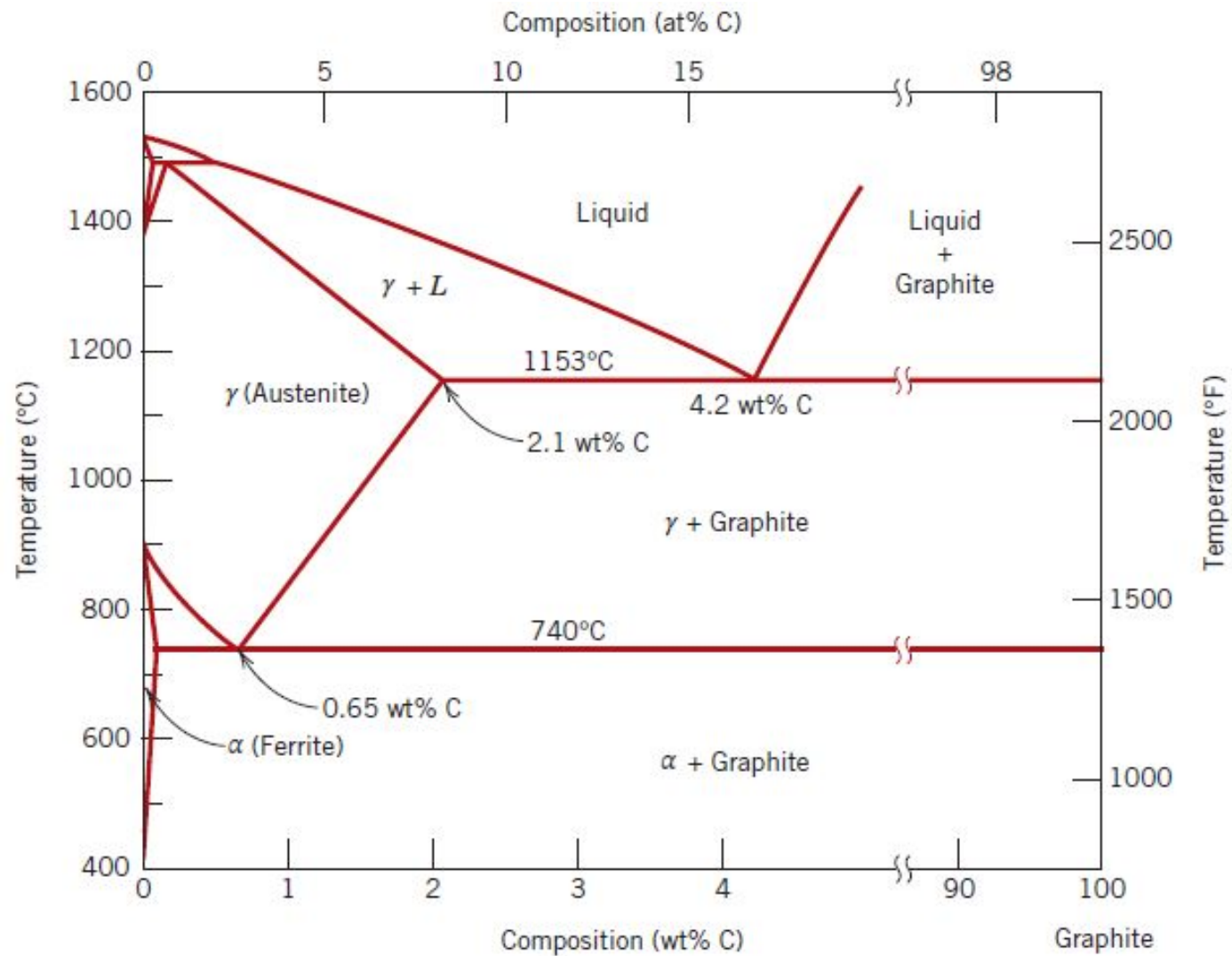
- Cementite (Fe_3C) is a metastable compound.
- Under certain conditions, it can decompose to form α -ferrite and graphite.



5. Equilibrium Diagram:

- The true equilibrium diagram for iron and carbon is represented in the figure next slide.
- Extends to 100 wt% C, with graphite as the carbon-rich phase, unlike cementite at 6.7 wt% C.

CAST IRONS...



CAST IRONS...

6. **Graphite Formation:**

- Regulated by composition and cooling rate.
- Presence of silicon in concentrations greater than about 1 wt% promotes graphite formation.
- Slower cooling rates during solidification favour the formation of graphite.

7. **Microstructure and Mechanical Behavior:**

- For most cast irons, carbon exists as graphite.
- Microstructure and mechanical behaviour depend on composition and heat treatment.

8. **Common Cast Iron Types:**

- Gray cast iron
- Nodular cast iron
- White cast iron
- Malleable cast iron
- Compacted graphite cast iron

GREY CAST IRONS...

1. Carbon and Silicon Contents:

- I. Carbon contents: Vary between 2.5 and 4.0 wt%.
- II. Silicon contents: Typically range between 1.0 and 3.0 wt%.

2. Graphite Form and Microstructure:

- I. Graphite exists in the form of flakes, resembling corn flakes.
- II. Typically surrounded by an α -ferrite or pearlite matrix.
- III. Microstructure depicted in the figure.

3. Fractured Surface and Appearance:

- I. Fractured surface appears grey due to the presence of graphite flakes.
- II. The graphite flakes have sharp and pointed tips, potentially causing stress concentration in external tensile stress applications.

4. Mechanical Properties:

- I. Comparatively weak and brittle in tension.
- II. Higher strength and ductility under compressive loads.
- III. Table 11.5 provides typical mechanical properties and compositions of common grey cast irons.

GREY CAST IRONS

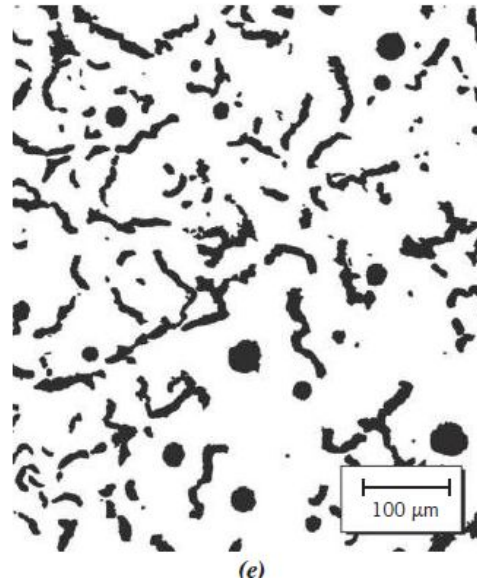
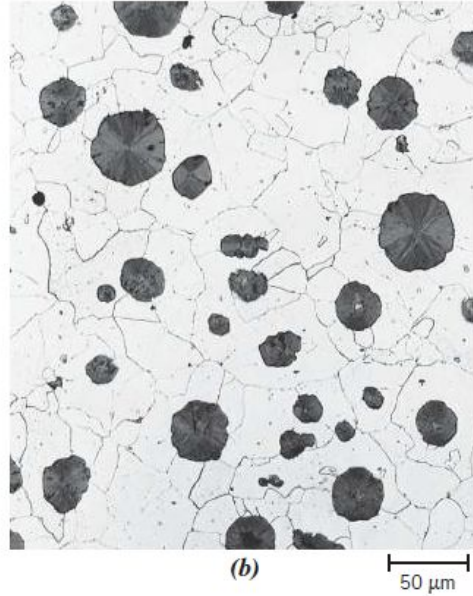
5. **Applications and Characteristics:**

- I. Effective in damping vibrational energy, making them suitable for structures exposed to vibrations.
- II. High resistance to wear.
- III. High fluidity in the molten state allows for casting intricate shapes with low shrinkage.
- IV. Cost-effective and among the least expensive metallic materials.

6. **Microstructure Modification:**

- I. Adjusting composition and/or using appropriate treatment can generate microstructures different from the typical one.
- II. Lowering silicon content or increasing cooling rate may alter microstructure, with graphite flakes embedded in a pearlite matrix.

GREY CAST IRONS



Optical photomicrographs
Of various cast irons.

(a) Gray iron: the dark graphite flakes are embedded in an α -ferrite matrix. 500 $\times$.

(b) Nodular (ductile) iron: the dark graphite nodules are surrounded by an α -ferrite matrix. 200 $\times$.

(c) White iron: the light cementite regions are surrounded by pearlite, which has the ferrite–cementite layered structure. 400 $\times$.

(d) Malleable iron: dark graphite rosettes (temper carbon) in an α -ferrite matrix. 150 $\times$.

(e) Compacted graphite iron: dark graphite wormlike particles are embedded within an α -ferrite matrix. 100 $\times$.

DUCTILE (OR NODULAR)

IRON

1. Microstructure Modification:

- I. Addition of a small amount of magnesium and/or cerium to grey iron before casting.
- II. Results in a different microstructure and set of mechanical properties.

2. Graphite Formation:

- I. Graphite forms as nodules or spherelike particles instead of flakes.

3. Alloy Name:

- I. Alloy produced is known as ductile or nodular iron.

4. Microstructure Characteristics:

- I. Matrix phase surrounding particles is either pearlite or ferrite, depending on heat treatment.
- II. Typically pearlite for as-cast pieces, but heat treatment at about 700°C (1300°F) yields a ferrite matrix.

5. Strength and Ductility:

- I. Castings are stronger and much more ductile compared to grey iron.
- II. Mechanical properties approach those of steel.
- III. Tensile strengths range between 380 and 480 MPa (55,000 and 70,000 psi).
- IV. Ductilities (percent elongation) range from 10% to 20%.

6. Applications:

- I. Used in various applications such as valves, pump bodies, crankshafts, gears, and other automotive and machine components.

WHITE CAST IRON

1. Composition Characteristics:

- Low-silicon cast irons (< 1.0 wt% Si) with rapid cooling rates.

2. Carbon Existence and Microstructure:

- Most carbon exists as cementite.
- Fracture surface appears white, termed "white cast iron."

3. Sectional Variation:

- Thick sections may have a surface layer of white iron, while interior regions cool more slowly, forming grey iron.

4. Mechanical Properties:

- Extremely hard due to significant cementite.
- Virtually unmachinable and very brittle.

5. Applications:

- Limited use for very hard and wear-resistant surfaces.
- Commonly found in applications like rollers in rolling mills.

6. Use in Production:

- Often used as an intermediary in the production of malleable iron.

7. Photomicrograph:

- Figure 11.3c illustrates the optical photomicrograph of white iron.

8. Relation to Malleable Iron:

Serves as a precursor in the production of malleable iron.

MALLEABLE CAST IRON

1. Malleable Iron Production:

- Heating white iron at 800-900°C (1470-1650°F) for an extended time in a neutral atmosphere decomposes cementite.
- Graphite forms in clusters or rosettes, surrounded by a ferrite or pearlite matrix

2. Microstructure and Photomicrograph:

- Ferritic malleable iron microstructure resembles nodular iron.
- Offers high strength and appreciable ductility or malleability.
- Photomicrograph displayed in Figure.

3. Mechanical Characteristics:

- Table lists typical mechanical characteristics.
- Relatively high strength and ductility.

4. Applications:

- Used in automotive components like connecting rods, transmission gears, and differential cases.
- Also employed in heavy-duty applications such as flanges, pipe fittings, and valve parts for railroad and marine services.

5. Production Comparison:

- Grey and ductile cast irons are produced in similar amounts.
- White and malleable cast irons produced in smaller quantities.

COMPACTED GRAPHITE IRON

1. Introduction to Compacted Graphite Iron (CGI):

- Relatively recent addition to the cast iron family, abbreviated as CGI.
- Carbon exists as graphite, promoted by silicon (1.7-3.0 wt%) with carbon concentration between 3.1 and 4.0 wt%.

2. Microstructure:

- Graphite in CGI alloys has a wormlike shape, intermediate between grey iron and ductile iron.
- Some graphite may exist as nodules (<20%) to avoid sharp edges, detrimental to fracture and fatigue resistance.

3. Additives:

- Magnesium and/or cerium are added in lower concentrations compared to ductile iron.
- Complex chemistries controlled for desired microstructure.

4. Matrix Phase and Heat Treatment:

- Matrix phase can be pearlite and/or ferrite depending on heat treatment.

COMPACTED GRAPHITE IRON

5. Mechanical Properties:

- Strength and ductility linked to microstructure and graphite particle shape.
- Degree of nodularity influences strength and ductility enhancements.
- CGIs with ferritic matrices have lower strengths but higher ductilities.

6. Comparisons with Other Cast Irons:

- Tensile and yield strengths comparable to ductile and malleable irons, greater than higher-strength grey irons.
- Ductilities intermediate between grey and ductile irons.
- Moduli of elasticity range between 140 and 165 GPa.

7. Desirable Characteristics of CGIs:

- Higher thermal conductivity.
- Better resistance to thermal shock.
- Lower oxidation at elevated temperatures.

8. Applications:

- Used in diesel engine blocks, exhaust manifolds, gearbox housings, brake discs for high-speed trains, and flywheels.

Table 11.5 Designations, Minimum Mechanical Properties, Approximate Compositions, and Typical Applications for Various Gray, Nodular, Malleable, and Compacted Graphite Cast Irons

Grade	UNS Number	Composition (wt%) ^a	Matrix Structure	Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Gray Iron							
SAE G1800	F10004	3.40–3.7 C, 2.55 Si, 0.7 Mn	Ferrite + pearlite	124 (18)	—	—	Miscellaneous soft iron castings in which strength is not a primary consideration
SAE G2500	F10005	3.2–3.5 C, 2.20 Si, 0.8 Mn	Ferrite + pearlite	173 (25)	—	—	Small cylinder blocks, cylinder heads, pistons, clutch plates, transmission cases
SAE G4000	F10008	3.0–3.3 C, 2.0 Si, 0.8 Mn	Pearlite	276 (40)	—	—	Diesel engine castings, liners, cylinders, and pistons
Ductile (Nodular) Iron							
ASTM A536 60–40–18	F32800	3.5–3.8 C, 2.0–2.8 Si, 0.05 Mg, <0.20 Ni, <0.10 Mo	Ferrite	414 (60)	276 (40)	18	Pressure-containing parts such as valve and pump bodies
100–70–03	F34800		Pearlite	689 (100)	483 (70)	3	High-strength gears and machine components
120–90–02	F36200		Tempered martensite	827 (120)	621 (90)	2	Pinions, gears, rollers, slides

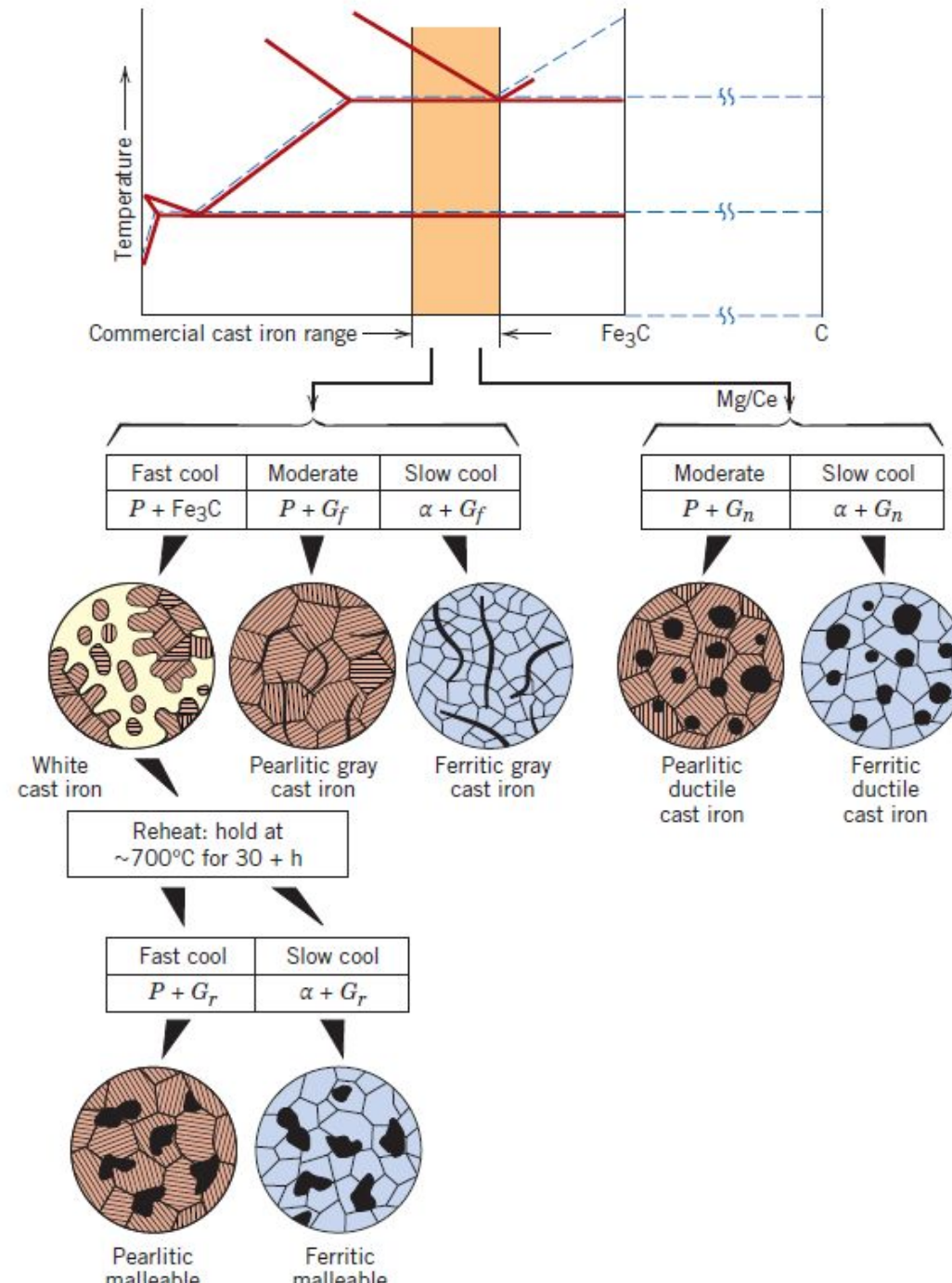
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Grade	UNS Number	Composition (wt%) ^a	Matrix Structure	Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Malleable Iron							
32510	F22200	2.3–2.7 C, 1.0–1.75 Si, <0.55 Mn	Ferrite	345 (50)	224 (32)	10	General engineering service at normal and elevated temperatures
45006	F23131	2.4–2.7 C, 1.25–1.55 Si, <0.55 Mn	Ferrite + pearlite	448 (65)	310 (45)	6	
Compacted Graphite Iron							
ASTM A842 Grade 250	—	3.1–4.0 C, 1.7–3.0 Si,	Ferrite	250 (36)	175 (25)	3	Diesel engine blocks, exhaust manifolds, brake discs for high-speed trains
Grade 450	—	0.015–0.035 Mg, 0.06–0.13 Ti	Pearlite	450 (65)	315 (46)	1	

^aThe balance of the composition is iron.

Source: Adapted from *ASM Handbook*, Vol. 1, *Properties and Selection: Irons, Steels, and High-Performance Alloys*, 1990. Reprinted by permission of ASM International, Materials Park, OH.

Figure 11.5 From the iron–carbon phase diagram, composition ranges for commercial cast irons. Also shown are schematic microstructures that result from a variety of heat treatments. G_f , flake graphite; G_r , graphite rosettes; G_n , graphite nodules; P , pearlite; α , ferrite.
(Adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, John Wiley & Sons, 1964. Reproduced with permission of Janet M. Moffatt.)



UNIFIED NUMBERING SYSTEM (UNS) FOR

STEEL:

1. **Definition:**
 - The Unified Numbering System (UNS) is a system developed by ASTM (American Society for Testing and Materials) and SAE (Society of Automotive Engineers) to uniquely identify and classify metals and alloys.
2. **Format:**
 - UNS designation for steel typically consists of a letter prefix followed by a series of numbers.
 - The letter prefix indicates the major category or alloy group, while the numeric code provides specific details about the steel composition.
3. **Letter Prefixes:**
 - **Letter "A":** Indicates a ferrous material (iron-based).
 - **Letter "G":** Represents AISI (American Iron and Steel Institute) and SAE carbon and alloy steels.
 - **Letter "S":** Refers to stainless steel.
 - Other letters may be used for specific alloy groups.
4. **Numeric Code:**
 - The numeric code following the letter prefix provides detailed information about the steel composition.
 - For carbon and alloy steels (prefix "G"), the first two digits represent the major alloying elements or, in some cases, carbon content.
5. **Example:**
 - UNS G10450 represents a carbon steel (G) with approximately 0.45% carbon content (10450).

UNIFIED NUMBERING SYSTEM (UNS) FOR STEEL:

6. **Special Codes:**

- Additional letters and numbers may be added to the UNS designation to specify certain details such as heat treatment, product form, or variations in chemical composition.

7. **Standardization:**

- UNS provides a standardized way of identifying steels, ensuring consistency in communication and specification across industries globally.

8. **Application:**

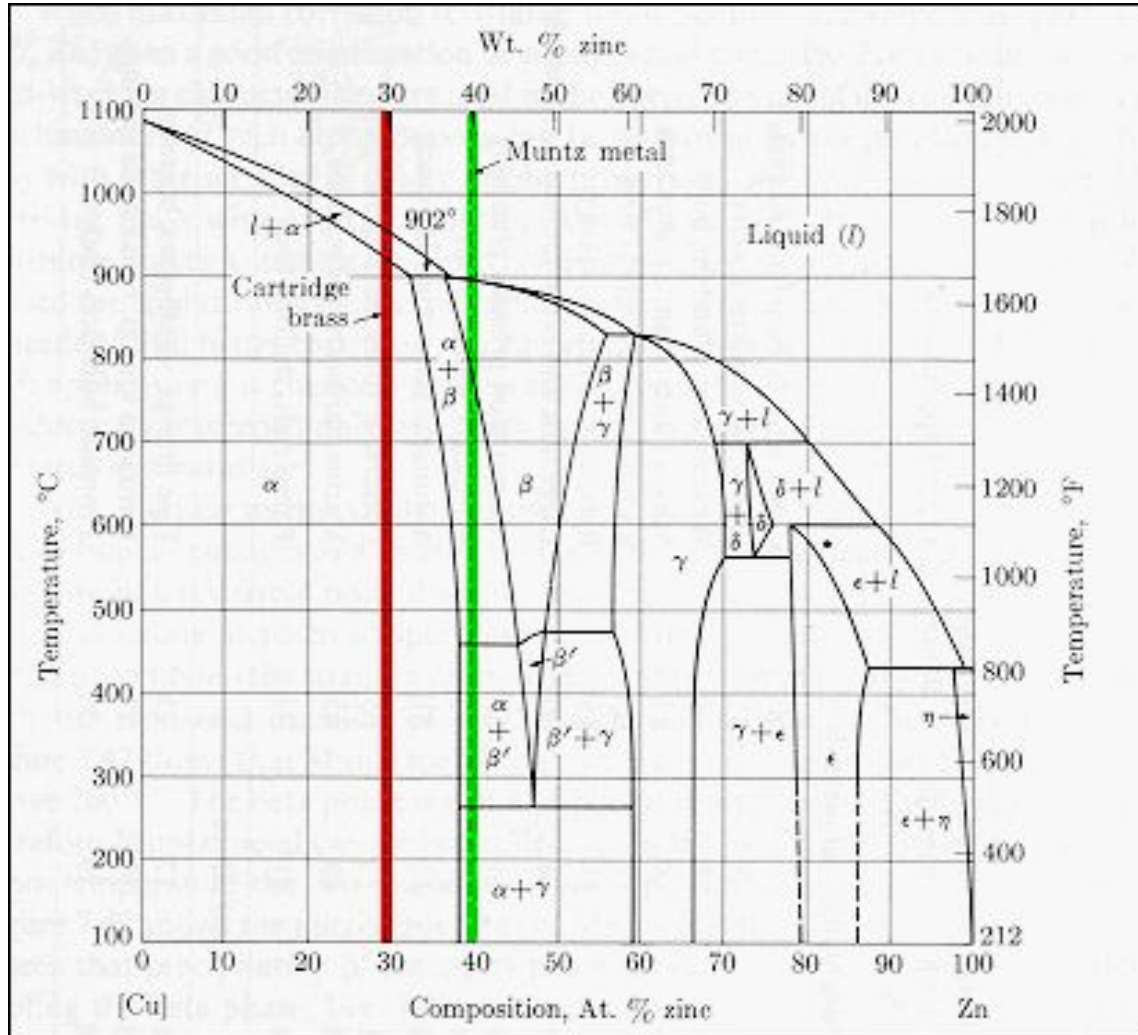
- Used in various industries including construction, automotive, aerospace, and manufacturing.
- Facilitates accurate material identification for design, manufacturing, and quality control purposes.

Understanding the UNS designation for steel is crucial for professionals in the materials and manufacturing industries to ensure proper material selection and compatibility.

COPPER AND ITS ALLOYS...

- **Historical Significance:** Copper and its alloys have been utilised across diverse applications since ancient times due to their unique physical properties.
- **Properties of Unalloyed Copper:** Softness and ductility hinder machining but enable extensive cold working; high corrosion resistance in environments like seawater and industrial settings.
- **Improvement through Alloying:** Alloying enhances mechanical and corrosion-resistance properties; heat treatments don't harden most copper alloys, necessitating cold working or solid-solution alloying.
- **Brasses – Common Copper Alloys:** Predominantly zinc-based, exhibit various phases based on zinc content; α phase (up to 35 wt% Zn) is soft and ductile, while higher zinc content introduces the harder β' phase.
- **Types of Brasses and Uses:** Yellow, naval, cartridge brass, muntz metal, gilding metal; applications span costume jewellery, cartridge casings, automotive radiators, musical instruments, electronics, and coins.
- **Bronzes – Copper Alloys with Other Elements:** Alloyed with tin, aluminium, silicon, and nickel for improved strength and corrosion resistance; ideal when both properties are crucial.
- **Heat-Treatable Copper Alloys – Beryllium Coppers:** Exceptional tensile strengths (up to 1400 MPa), excellent electrical properties, corrosion resistance, and wear resistance with proper lubrication; achieved through precipitation-hardening heat treatments, albeit at a

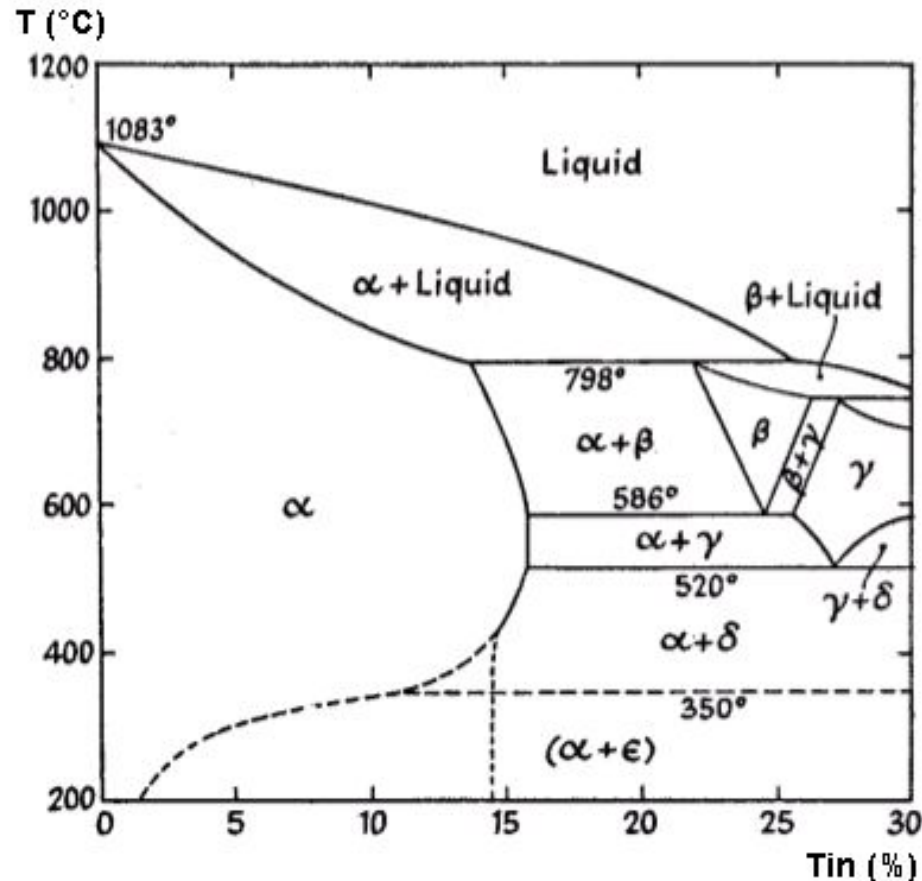
COPPER AND ITS ALLOYS...



Brass

- The [Copper-Zinc](#) binary phase diagram ([brass](#)) is another example of a complex phase diagram which shows many invariant reactions.
- Two common alloy compositions are shown, Cartridge brass Cu-30 wt % Zn, Muntz metal (green) Cu-40 wt % Zn.

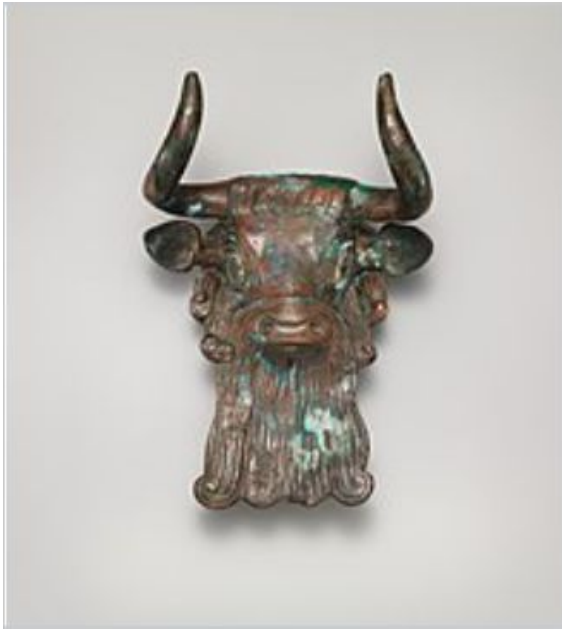
COPPER AND ITS ALLOYS...



Bronze compositions vary widely based on the specific application and desired properties. However, some common bronze **compositions** include:

- **Copper:** Typically forms the majority of the alloy, often comprising around **80-95%** of the bronze composition.
- **Tin:** The addition of tin, usually in the range of **5-20%**, is a defining characteristic of bronze. Higher tin content tends to increase hardness and strength.
- **Other Alloying Elements:** Depending on the desired properties, other elements like aluminum, silicon, nickel, or phosphorus might be added in smaller quantities (usually less than 5%) to modify characteristics such as strength, corrosion resistance, and machinability.

Bronze Artwork



Brass Artifacts



Bronze typically contains **tin** as a primary alloying element, which can be costlier than **zinc**, the primary alloying element in **brass**. Additionally, other elements used in bronze, such as aluminum or nickel, might affect the cost compared to brass compositions.

Compositions, Mechanical Properties, and Typical Applications for Eight Copper Alloys

Table 11.6 Compositions, Mechanical Properties, and Typical Applications for Eight Copper Alloys

Alloy Name	UNS Number	Composition (wt%) ^a	Condition	Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Wrought Alloys							
Electrolytic tough pitch	C11000	0.04 O	Annealed	220 (32)	69 (10)	45	Electrical wire, rivets, screening, gaskets, pans, nails, roofing
Beryllium copper	C17200	1.9 Be, 0.20 Co	Precipitation hardened	1140–1310 (165–190)	965–1205 (140–175)	4–10	Springs, bellows, firing pins, bushings, valves, diaphragms
Cartridge brass	C26000	30 Zn	Annealed	300 (44)	75 (11)	68	Automotive radiator cores, ammunition components, lamp fixtures, flashlight shells, kickplates
			Cold-worked (H04 hard)	525 (76)	435 (63)	8	
Phosphor bronze, 5% A	C51000	5 Sn, 0.2 P	Annealed Cold-worked (H04 hard)	325 (47) 560 (81)	130 (19) 515 (75)	64 10	Bellows, clutch disks, diaphragms, fuse clips, springs, welding rods
Copper–nickel, 30%	C71500	30 Ni	Annealed	380 (55)	125 (18)	36	Condenser and heat-exchanger components, saltwater piping
			Cold-worked (H02 hard)	515 (75)	485 (70)	15	
Cast Alloys							
Leaded yellow brass	C85400	29 Zn, 3 Pb, 1 Sn	As cast	234 (34)	83 (12)	35	Furniture hardware, radiator fittings, light fixtures, battery clamps
Tin bronze	C90500	10 Sn, 2 Zn	As cast	310 (45)	152 (22)	25	Bearings, bushings, piston rings, steam fittings, gears
Aluminum bronze	C95400	4 Fe, 11 Al	As cast	586 (85)	241 (35)	18	Bearings, gears, worms, bushings, valve seats and guards, pickling hooks

^aThe balance of the composition is copper.

Source: Adapted from *ASM Handbook*, Vol. 2, *Properties and Selection: Nonferrous Alloys and Special-Purpose Materials*, 1990. Reprinted by permission of ASM International, Materials Park, OH.

ALUMINIUM AND ITS ALLOYS...

1. Aluminum Characteristics:

- Low density compared to steel (2.7 g/cm^3 vs. 7.9 g/cm^3)
- High electrical and thermal conductivities
- Resistance to corrosion in various environments, including ambient atmosphere
- High ductility, evidenced by thin aluminium foil production

2. Ductility and Crystal Structure:

- Maintains ductility due to FCC crystal structure even at low temperatures
- Limitation: Low melting temperature (660°C or 1220°F) restricts use at high temperatures

3. Enhancing Mechanical Strength:

- Strength enhancement through cold work and alloying
- Trade-off: Strength enhancement may decrease corrosion resistance
- Principal alloying elements: copper, magnesium, silicon, manganese, zinc

4. Alloy Classification and Designation:

- Classification as cast or wrought aluminium alloys
- Four-digit numbering system indicates impurities and purity levels
- Decimal point for cast alloys; hyphen followed by temper designation (e.g., F, H, O)
- Table 11.7 presents the temper designation scheme

ALUMINIUM AND ITS ALLOYS...

5. **Properties, Compositions, and Applications:**

- Table 11.8 details compositions, properties, and applications of various wrought and cast aluminium alloys
- Applications include aircraft structural parts, beverage cans, bus bodies, automotive components

6. **Aluminum Alloys for Fuel Efficiency:**

- Attention towards aluminium, magnesium, and titanium alloys for transportation to reduce fuel consumption
- Specific strength: Tensile strength—specific gravity ratio; enables load sustainability despite lower tensile strength

7. **Advancements: Aluminum-Lithium Alloys:**

- New generation alloys developed for aircraft/aerospace industries
- Low densities ($2.5\text{-}2.6\text{ g/cm}^3$), high specific moduli, excellent fatigue and low-temperature toughness
- Precipitation hardening is possible but more costly due to lithium's reactivity, requiring special processing techniques

ALUMINIUM AND ITS ALLOYS...

Table 11.7 Temper Designation Scheme for Aluminum Alloys

<i>Designation</i>	<i>Description</i>
Basic Tempers	
F	As-fabricated—by casting or cold working
O	Annealed—lowest strength temper (wrought products only)
H	Strain-hardened (wrought products only)
W	Solution heat-treated—used only on products that precipitation harden naturally at room temperature over periods of months or years
T	Solution heat-treated—used on products that strength stabilize within a few weeks—followed by one or more digits
Strain-Hardened Tempers^a	
H1	Strain-hardened only
H2	Strain-hardened and then partially annealed
H3	Strain-hardened and then stabilized
Heat-Treating Tempers^b	
T1	Cooled from an elevated-temperature shaping process and naturally aged
T2	Cooled from an elevated-temperature shaping process, cold worked, and naturally aged
T3	Solution heat-treated, cold worked, and naturally aged
T4	Solution heat-treated and naturally aged
T5	Cooled from an elevated-temperature shaping process and artificially aged
T6	Solution heat-treated and artificially aged
T7	Solution heat-treated and overaged or stabilized
T8	Solution heat-treated, cold worked, and artificially aged
T9	Solution heat-treated, artificially aged, and cold worked
T10	Cooled from an elevated-temperature shaping process, cold worked, and artificially aged

^aTwo additional digits may be added to denote degree of strain hardening.

^bAdditional digits (the first of which cannot be zero) are used to denote variations of these 10 tempers.

Source: Adapted from *ASM Handbook*, Vol. 2, *Properties and Selection: Nonferrous Alloys and Special-Purpose Materials*, 1990. Reproduced with permission of ASM International, Materials Park, OH, 44073.

ALUMINIUM AND ITS ALLOYS...

Table 11.8 Compositions, Mechanical Properties, and Typical Applications for Several Common Aluminum Alloys

Aluminum Association Number	UNS Number	Composition (wt%) ^a	Condition (Temper Designation)	Mechanical Properties			Typical Applications/ Characteristics
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Wrought, Non-Heat-Treatable Alloys							
1100	A91100	0.12 Cu	Annealed (O)	90 (13)	35 (5)	35–45	Food/chemical handling and storage equipment, heat exchangers, light reflectors
3003	A93003	0.12 Cu, 1.2 Mn, 0.1 Zn	Annealed (O)	110 (16)	40 (6)	30–40	Cooking utensils, pressure vessels and piping
5052	A95052	2.5 Mg, 0.25 Cr	Strain hardened (H32)	230 (33)	195 (28)	12–18	Aircraft fuel and oil lines, fuel tanks, appliances, rivets, and wire
Wrought, Heat-Treatable Alloys							
2024	A92024	4.4 Cu, 1.5 Mg, 0.6 Mn	Heat-treated (T4)	470 (68)	325 (47)	20	Aircraft structures, rivets, truck wheels, screw machine products
6061	A96061	1.0 Mg, 0.6 Si, 0.30 Cu, 0.20 Cr	Heat-treated (T4)	240 (35)	145 (21)	22–25	Trucks, canoes, railroad cars, furniture, pipelines
7075	A97075	5.6 Zn, 2.5 Mg, 1.6 Cu, 0.23 Cr	Heat-treated (T6)	570 (83)	505 (73)	11	Aircraft structural parts and other highly stressed applications

ALUMINIUM AND ITS ALLOYS...

Table 11.8 Compositions, Mechanical Properties, and Typical Applications for Several Common Aluminum Alloys

Aluminum Association Number	UNS Number	Composition (wt%) ^a	Condition (Temper Designation)	Mechanical Properties			Typical Applications/ Characteristics
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Cast, Heat-Treatable Alloys							
295.0	A02950	4.5 Cu, 1.1 Si	Heat-treated (T4)	221 (32)	110 (16)	8.5	Flywheel and rear-axle housings, bus and aircraft wheels, crankcases
356.0	A03560	7.0 Si, 0.3 Mg	Heat-treated (T6)	228 (33)	164 (24)	3.5	Aircraft pump parts, automotive transmission cases, water-cooled cylinder blocks
Aluminum–Lithium Alloys							
2090	—	2.7 Cu, 0.25 Mg, 2.25 Li, 0.12 Zr	Heat-treated, cold-worked (T83)	455 (66)	455 (66)	5	Aircraft structures and cryogenic tankage structures
8090	—	1.3 Cu, 0.95 Mg, 2.0 Li, 0.1 Zr	Heat-treated, cold-worked (T651)	465 (67)	360 (52)	—	Aircraft structures that must be highly damage tolerant

^aThe balance of the composition is aluminum.

Source: Adapted from *ASM Handbook*, Vol. 2, *Properties and Selection: Nonferrous Alloys and Special-Purpose Materials*, 1990. Reprinted by permission of ASM International, Materials Park, OH.

MAGNESIUM AND ITS ALLOYS

1. Magnesium's Key Characteristics:

- Lowest density among structural metals (1.7 g/cm^3), ideal for weight-sensitive applications like aircraft components
- HCP crystal structure, relatively soft, with a low elastic modulus (45 GPa)

2. Deformation and Fabrication Challenges:

- Difficult to deform at room temperature; limited cold work possible without annealing
- Fabrication primarily through casting or hot working ($200^\circ\text{C} - 350^\circ\text{C}$ or $400^\circ\text{F} - 650^\circ\text{F}$)
- Moderately low melting temperature (651°C or 1204°F)

3. Corrosion and Chemical Properties:

- Relatively unstable, susceptible to corrosion in marine environments
- Reasonable resistance to corrosion or oxidation in normal atmospheric conditions due to impurities

4. Classification and Alloying Elements:

- Classified as cast or wrought, some heat-treatable
- Major alloying elements: aluminum, zinc, manganese, rare earth

5. Common Magnesium Alloys:

- Table 11.9 details the compositions, properties, and applications of several magnesium alloys
- Usage in aircraft, missile applications, luggage, and various industries

6. Increasing Demand and Applications:

- Recent surge in demand across diverse industries
- Replacing engineering plastics due to stiffness, recyclability, and lower production costs
- Applications in handheld devices, automobiles (steering wheels, transmission cases), audio-video equipment, and communication devices.

MAGNESIUM AND ITS ALLOYS

Table 11.9 Compositions, Mechanical Properties, and Typical Applications for Six Common Magnesium Alloys

ASTM Number	UNS Number	Composition (wt%) ^a	Condition	Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Wrought Alloys							
AZ31B	M11311	3.0 Al, 1.0 Zn, 0.2 Mn	As extruded	262 (38)	200 (29)	15	Structures and tubing, cathodic protection
HK31A	M13310	3.0 Th, 0.6 Zr	Strain hardened, partially annealed	255 (37)	200 (29)	9	High strength to 315°C (600°F)
ZK60A	M16600	5.5 Zn, 0.45 Zr	Artificially aged	350 (51)	285 (41)	11	Forgings of maximum strength for aircraft
Cast Alloys							
AZ91D	M11916	9.0 Al, 0.15 Mn, 0.7 Zn	As cast	230 (33)	150 (22)	3	Die-cast parts for automobiles, luggage, and electronic devices
AM60A	M10600	6.0 Al, 0.13 Mn	As cast	220 (32)	130 (19)	6	Automotive wheels
AS41A	M10410	4.3 Al, 1.0 Si, 0.35 Mn	As cast	210 (31)	140 (20)	6	Die castings requiring good creep resistance

^aThe balance of the composition is magnesium.

Source: Adapted from *ASM Handbook*, Vol. 2, *Properties and Selection: Nonferrous Alloys and Special-Purpose Materials*, 1990. Reprinted by permission of ASM International, Materials Park, OH.

TITANIUM AND ITS ALLOYS

•Titanium's Unique Properties:

- Low density (4.5 g/cm³), high melting point (1668°C or 3035°F), elastic modulus of 107 GPa. Titanium alloys exhibit exceptional strength (up to 1400 MPa), yielding remarkable specific strengths
- High ductility, easy forging, and machining capabilities

•Crystal Structure and Alloy Phases:

- Unalloyed titanium: Hexagonal close-packed (α phase) at room temperature; transforms to body-centred cubic (β phase) at 883°C (1621°F)
- Alloying elements like vanadium, niobium, and molybdenum influence phase stability and transformation temperatures

•Titanium Alloy Classifications:

- α , β , $\alpha + \beta$, and near- α alloys based on the dominant phase(s) after processing

•Characteristics of Alloy Types:

- α -titanium alloys (alloyed with aluminium and tin) preferred for high-temperature applications, limited forgeability
- β -titanium alloys exhibit high forgeability and high fracture toughness, retaining the β phase at room temperature

• $\alpha + \beta$ and Near- α Alloys:

- $\alpha + \beta$ alloys allow for improved strength through heat treatment, diverse microstructures, and good formability
- Near- α alloys have limited β content but similar properties to α alloys with a broader range of microstructures

TITANIUM AND ITS ALLOYS

Table 11.10 Compositions, Mechanical Properties, and Typical Applications for Several Common Titanium Alloys

Alloy Type	Common Name (UNS Number)	Composition (wt%)	Condition	Average Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Commercially pure	Unalloyed (R50250)	99.5 Ti	Annealed	240 (35)	170 (25)	24	Jet engine shrouds, cases and airframe skins, corrosion-resistant equipment for marine and chemical processing industries
α	Ti-5Al-2.5Sn (R54520)	5 Al, 2.5 Sn, balance Ti	Annealed	826 (120)	784 (114)	16	Gas turbine engine casings and rings; chemical processing equipment requiring strength to temperatures of 480°C (900°F)
Near α	Ti-8Al-1Mo-1V (R54810)	8 Al, 1 Mo, 1 V, balance Ti	Annealed (duplex)	950 (138)	890 (129)	15	Forgings for jet engine components (compressor disks, plates, and hubs)
$\alpha + \beta$	Ti-6Al-4V (R56400)	6 Al, 4 V, balance Ti	Annealed	947 (137)	877 (127)	14	High-strength prosthetic implants, chemical-processing equipment, airframe structural components
$\alpha + \beta$	Ti-6Al-6V-2Sn (R56620)	6 Al, 2 Sn, 6 V, 0.75 Cu, balance Ti	Annealed	1050 (153)	985 (143)	14	Rocket engine case airframe applications and high-strength airframe structures
β	Ti-10V-2Fe-3Al	10 V, 2 Fe, 3 Al, balance Ti	Solution + aging	1223 (178)	1150 (167)	10	Best combination of high strength and toughness of any commercial titanium alloy; used for applications requiring uniformity of tensile properties at surface and center locations; high-strength airframe components

Source: Adapted from *ASM Handbook*, Vol. 2, *Properties and Selection: Nonferrous Alloys and Special-Purpose Materials*, 1990. Reprinted by permission of ASM International, Materials Park, OH.

CERAMICS...

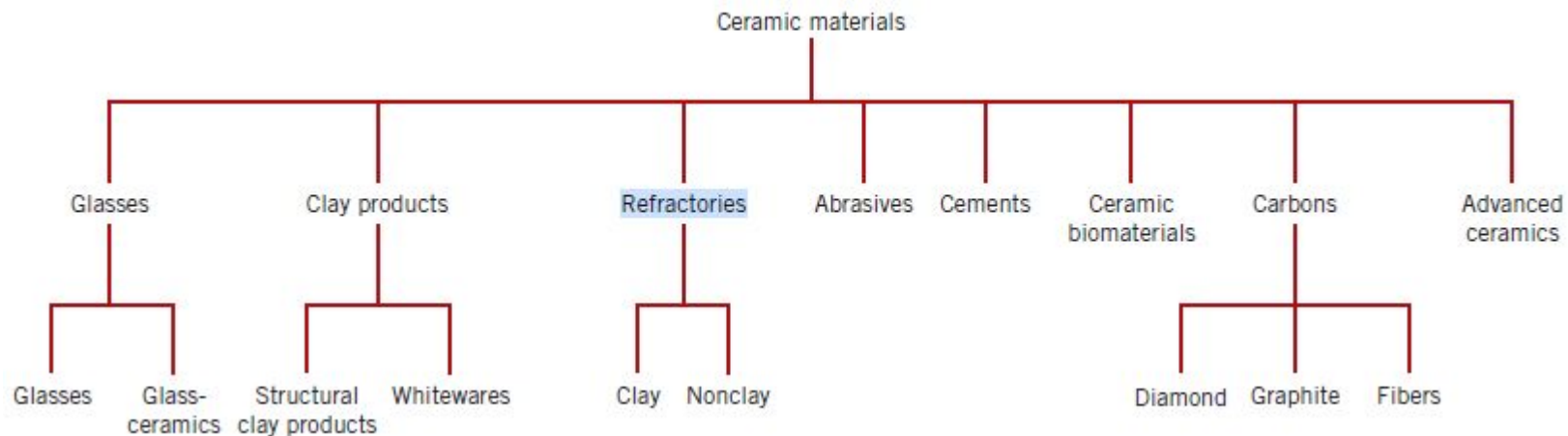
Ceramics are a class of materials characterized by their inorganic nature and a combination of metallic and non-metallic elements. In contrast to metals, ceramics exhibit a significant disparity in physical characteristics, leading to distinct applications. This diversity in properties allows ceramics to complement other materials, such as metals and polymers, in various industrial and technological applications.

The application–classification scheme for ceramic materials encompasses several groups, each with unique properties and uses:

1. **Glasses:** Amorphous solids with a random atomic arrangement, glasses are transparent and often used in windows, containers, and optical devices.
2. **Structural Clay Products:** This group includes bricks, tiles, and other clay-based materials used for construction purposes due to their strength and durability.
3. **Whitewares:** These are ceramics with a white or light-colored appearance, such as porcelain and fine China, commonly used for decorative and functional purposes.
4. **Refractories:** Heat-resistant ceramics are employed in high-temperature applications like furnaces, kilns, and industrial processes.

CERAMICS...

5. **Abrasives:** Ceramics are used as abrasive materials for cutting, grinding, and polishing applications due to their hardness and wear resistance.
6. **Cements:** Ceramic materials that, when mixed with water, form a solid mass, widely utilized in construction for bonding and structural purposes.
7. **Ceramic Biomaterials:** Ceramics designed for medical applications, including dental implants and bone substitutes, due to their biocompatibility and resistance to corrosion.
8. **Carbons:** Carbon-based ceramics with unique properties, often used in electronic components, high-temperature applications, and as structural materials.
9. **Advanced Ceramics:** Newly developed ceramics with enhanced properties, such as high strength, toughness, and thermal resistance, expanding their applications in cutting-edge techr



CERAMICS: GLASSES

Glasses, a well-recognized subset of ceramics, find widespread use in everyday items such as containers, lenses, and fibreglass. These materials, characterized as noncrystalline silicates, incorporate various oxides, including CaO, Na₂O, K₂O, and Al₂O₃, which play a significant role in determining the properties of glass. A standard soda–lime glass, for example, comprises approximately 70 wt% SiO₂, with the remaining portion primarily constituted by Na₂O (soda) and CaO (lime). The standout features of these materials include their optical transparency and the notable

Table 13.1 Compositions and Characteristics of Some Common Commercial Glasses

<i>Glass Type</i>	<i>Composition (wt%)</i>						<i>Characteristics and Applications</i>
	<i>SiO₂</i>	<i>Na₂O</i>	<i>CaO</i>	<i>Al₂O₃</i>	<i>B₂O₃</i>	<i>Other</i>	
Fused silica	>99.5						High melting temperature, very low coefficient of expansion (thermally shock-resistant)
96% Silica (Vycor)	96				4		Thermally shock- and chemically resistant—laboratory ware
Borosilicate (Pyrex)	81	3.5		2.5	13		Thermally shock- and chemically resistant—ovenware
Container (soda–lime)	74	16	5	1		4 MgO	Low melting temperature, easily worked, also durable
Fiberglass	55		16	15	10	4 MgO	Easily drawn into fibers—glass–resin composites
Optical flint	54	1				37 PbO, 8 K ₂ O	High density and high index of refraction—optical lenses
Glass-ceramic (Pyroceram)	43.5	14		30	5.5	6.5 TiO ₂ , 0.5 As ₂ O ₃	Easily fabricated; strong; resists thermal shock—ovenware

GLASS – CERAMICS...

- Inorganic glasses can transform from a noncrystalline state to a crystalline state through high-temperature heat treatment, a process known as crystallization.
- This results in the formation of fine-grained polycrystalline materials referred to as glass-ceramics. The formation of these grains involves nucleation and growth stages, and the kinetics of crystallization can be described using principles similar to those applied to phase transformations in metal systems.
- The degree of transformation can be expressed with isothermal and continuous-cooling transformation diagrams. Figure 13.2 illustrates a continuous-cooling transformation diagram for the crystallization of lunar glass, showing similarities to eutectoid composition curves for iron–carbon alloys.
- Nucleating agents, often containing titanium dioxide, are added to promote crystallization, causing shifts in transformation curves to shorter times. Additionally, a critical cooling rate is identified, below which glass-ceramic material forms.

GLASS – CERAMICS...

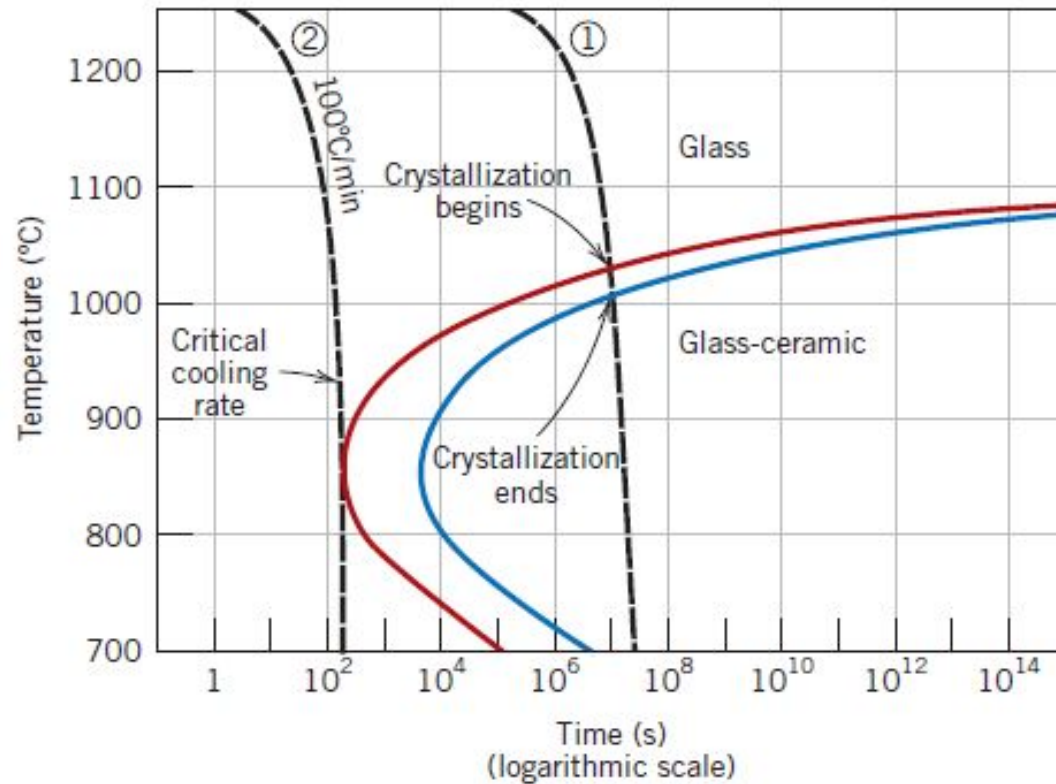


Figure 13.2 Continuous-cooling transformation diagram for the crystallization of a lunar glass (35.5 wt% SiO₂, 14.3 wt% TiO₂, 3.7 wt% Al₂O₃, 23.5 wt% FeO, 11.6 wt% MgO, 11.1 wt% CaO, and 0.2 wt% Na₂O). Superimposed on this plot are two cooling curves, labeled 1 and 2. (Reprinted from *Glass: Science and Technology*, Vol. 1, D. R. Uhlmann and N. J. Kreidl (Editors), "The Formation of Glasses," p. 22, copyright 1983, with permission from Elsevier.)

GLASS CERAMICS-Properties and Applications

1. **Glass-Ceramic Characteristics:**

- High mechanical strengths, low thermal expansion, good high-temperature capabilities, suitable dielectric properties, biologically compatible, varied transparency (opaque or transparent)

2. **Ease of Fabrication:**

- Conventional glass-forming techniques, mass production of nearly pore-free ware

3. **Commercial Trade Names:**

- Pyroceram, CorningWare, Cerdur, Vision

4. **Common Uses:**

- Ovenware, tableware, oven windows, range tops, electrical insulators, substrates for printed circuit boards, architectural cladding, heat exchangers and regenerators

5. **Included in Table 13.1:**

1. Typical glass-ceramic composition details

CLAY PRODUCTS

1. Clay as a Widely Used Ceramic Raw Material:

- Abundant and inexpensive, clay is a commonly used ceramic raw material.
- Often used in its natural state without quality upgrades.

2. Ease of Forming Clay Products:

- Clay, when mixed with water in proper proportions, forms a plastic mass, making it highly moldable.

3. Processing Steps:

Formed pieces are dried to reduce moisture. Elevated-temperature firing enhances mechanical strength.

4. Broad Classifications of Clay Products:

1. Structural Clay Products:

- Examples: Building bricks, tiles, sewer pipes.
- Emphasizes structural integrity in applications.

2. Whiteware Ceramics:

- Become white after high-temperature firing.
- Examples: Porcelain, pottery, tableware, china, plumbing fixtures (sanitary ware).

6. Composition and Influences:

Many clay-based products contain nonplastic ingredients influencing drying, firing processes, and final characteristics. Clays are aluminosilicates composed of alumina (Al_2O_3) and silica (SiO_2) that contain chemically bound water. They have a broad range of physical characteristics, chemical compositions, and structures; common impurities include compounds (usually oxides) of barium, calcium, sodium, potassium, iron, and also some organic matter. Crystal structures for clay minerals are relatively complicated; however, one prevailing characteristic is a layered structure. The most common clay minerals that are of interest have what is called the kaolinite structure $[\text{Al}_2(\text{Si}_2\text{O}_5)(\text{OH})_4]$.

REFRACTORIES

Clay Refractories: Subclassified into two categories- fireclay and high-alumina.

1. Fireclay Refractories:

Primary ingredients: High-purity fireclays with alumina and silica mixtures (25-45 wt% alumina).

According to the $\text{SiO}_2\text{--Al}_2\text{O}_3$ phase diagram (Figure 12.25), the highest temperature without a liquid phase is 1587°C (2890°F). Mullite and silica (cristobalite) are equilibrium phases below this temperature.

The presence of a small amount of liquid phase during service use is permissible without compromising mechanical integrity. Above 1587°C , the fraction of the liquid phase depends on refractory composition.

Higher alumina content increases maximum service temperature, allowing for a small amount of liquid formation.

2. High-Alumina Refractories:

Principal ingredient: Bauxite, a mineral composed of aluminium hydroxide $\text{Al}(\text{OH})_3$ and kaolinite clays.

Alumina content varies between 50 and 87.5 wt%. More robust at high temperatures compared to fireclay refractories. Suited for exposure to more severe environments.

SUPERALLOYS...

Definition: Superalloys are a group of high-performance materials designed for extreme temperature, pressure, and corrosive environments.

Key Characteristics

- 1.Heat Resistance:** Maintain strength at high temperatures, often above 1000°C.
- 2.Corrosion Resistance:** Excellent resistance to corrosion and oxidation.
- 3.Strength and Durability:** Exceptional mechanical strength and durability even in harsh conditions.
- 4.Versatility:** Widely used in aerospace, gas turbine engines, chemical processing, and nuclear reactors.
- 5.Composition:** Base Metals- Nickel, cobalt, or iron-based. Alloying Elements- Chromium, aluminium, titanium, and others for enhanced properties.

SUPERALLOYS...

6. Applications

- **Aerospace Industry:** Jet engines, turbine blades, and other high-temperature components.
- **Energy Sector:** Gas turbine power plants, nuclear reactors.
- **Chemical Processing:** Corrosion-resistant components for aggressive environments.
- **Medical Devices:** Biocompatible alloys for medical implants.

7. Examples

- **Inconel:** Used in aircraft components and gas turbine engines.
- **Hastelloy:** Resistant to corrosion and used in chemical processing.
- **Waspaloy:** High-temperature applications, such as turbine blades.

8. Challenges

- **Cost:** High manufacturing costs due to complex alloy compositions and processing.
- **Processing Difficulty:** Challenges in casting and forming due to high melting points.

SUPERALLOYS

Table 11.11 Compositions for Several Superalloys

Alloy Name	Composition (wt%)									
	Ni	Fe	Co	Cr	Mo	W	Ti	Al	C	Other
Iron–Nickel (Wrought)										
A-286	26	55.2	—	15	1.25	—	2.0	0.2	0.04	0.005 B, 0.3 V
Incoloy 925	44	29	—	20.5	2.8	—	2.1	0.2	0.01	1.8 Cu
Nickel (Wrought)										
Inconel-718	52.5	18.5	—	19	3.0	—	0.9	0.5	0.08	5.1 Nb, 0.15 max Cu
Waspaloy	57.0	2.0 max	13.5	19.5	4.3	—	3.0	1.4	0.07	0.006 B, 0.09 Zr
Nickel (Cast)										
Rene 80	60	—	9.5	14	4	4	5	3	0.17	0.015 B, 0.03 Zr
Mar-M-247	59	0.5	10	8.25	0.7	10	1	5.5	0.15	0.015 B, 3 Ta, 0.05 Zr, 1.5 Hf
Cobalt (Wrought)										
Haynes 25 (L-605)	10	1	54	20	—	15	—	—	0.1	
Cobalt (Cast)										
X-40	10	1.5	57.5	22	—	7.5	—	—	0.50	0.5 Mn, 0.5 Si

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SHAPE-MEMORY ALLOYS (SMAS)...

1. **Definition:** A group of metals, including nickel-titanium alloys (Nitinol), exhibiting shape-memory phenomena.
2. **Shape-Memory Effect:**
 - Deforms at low temperatures.
 - Returns to original shape upon proper heat treatment.
 - Common alloys: Nitinol, Cu–Zn–Al, Cu–Al–Ni.
3. **Polymorphic Nature:**
 - Two crystal structures (austenite and martensite) involved in the shape-memory effect.
 - Deformation occurs through the migration of twin boundaries.
4. **Deformation Process:**
 - Deformation at martensite phase, retention of shape when stress is removed.
 - Subsequent heating reverts material to its original size and shape.

SHAPE-MEMORY ALLOYS (SMAS)...

6. Transformation Temperatures:

- Martensite-to-austenite transformation occurs between A_s (austenite start) and A_f (austenite finish) temperatures.
- Transformation cycle is repeatable.

7. Thermoelastic Property:

- Deformation is nonpermanent; termed thermoelastic.
- Heat treatment restores the original shape.

8. Applications:

- Aerospace (hydraulic lines, joints on pipelines, plumbing on ships and submarines).
- Other applications include eyeglass frames, tooth-straightening braces, collapsible antennas, greenhouse window openers, anti-scald control valves, women's foundation garments, fire sprinkler valves, and biomedical devices.

9. Versatility:

- Can vary transformation temperatures by adjusting Ni–Ti ratio and adding other elements.
- Falls under the category of "smart materials" due to sensing and responding to temperature changes.

SHAPE-MEMORY ALLOYS (SMAS)...

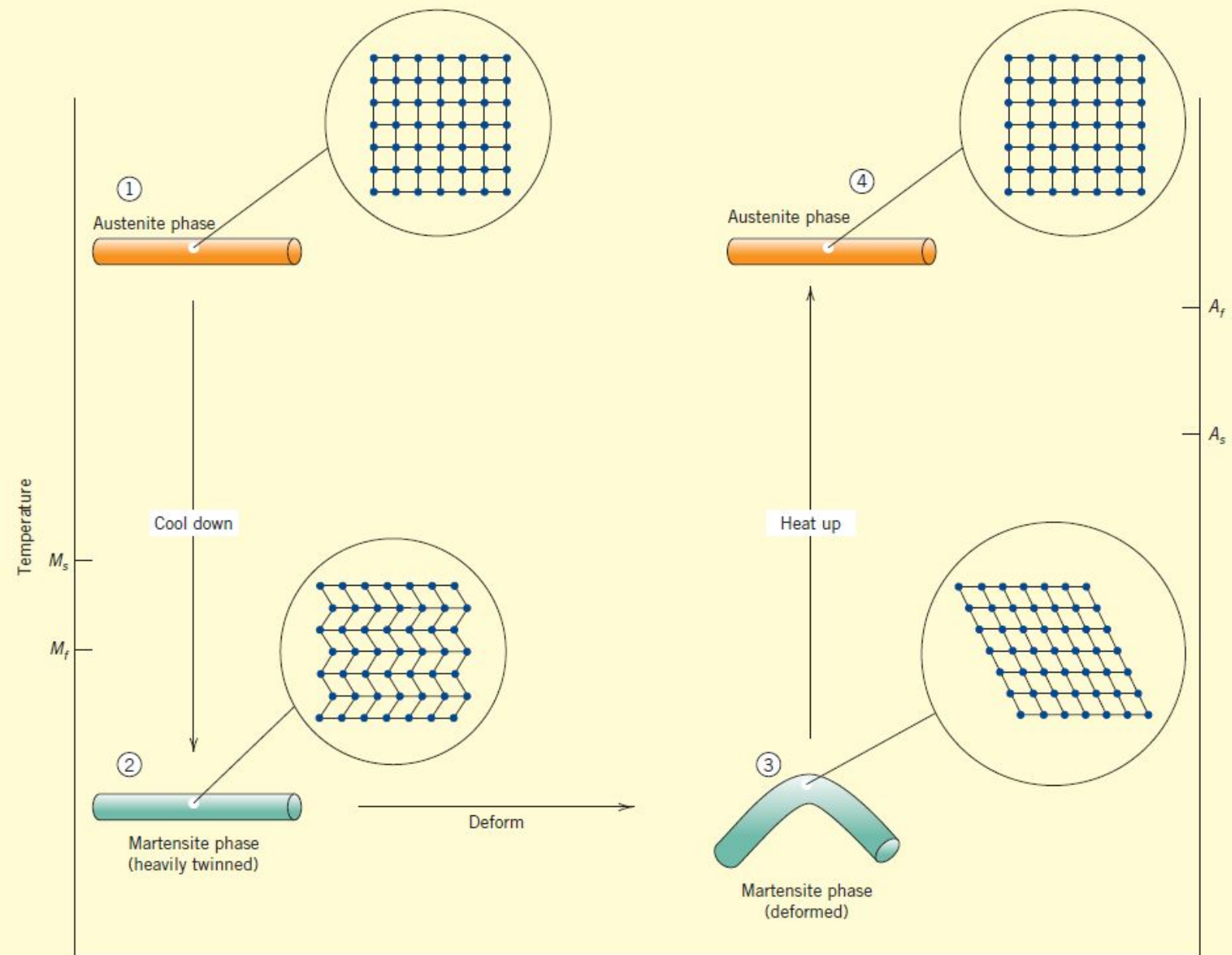


Figure 10.38 Diagram illustrating the shape-memory effect. The insets are schematic representations of the crystal structure at the four stages. M_s and M_f denote temperatures at which the martensitic transformation begins and ends, respectively. Likewise for the austenite transformation, A_s and A_f represent the respective beginning and end transformation temperatures.

NANOMATERIALS...

1. Definition:

- Recently discovered carbon materials with unique properties.
- Crucial in cutting-edge technologies and anticipated to have a significant role in future high-tech applications.

2. Three Key Nanocarbons:

- **Fullerenes:** Hollow carbon molecules.
- **Carbon Nanotubes:** Tubular structures.
- **Graphene:** Single-layer carbon atoms arranged in a hexagonal lattice.

3. Nano Scale:

- Particle size is less than about 100 nanometers.

4. Bonding Structure:

- Carbon atoms within each nanoparticle bonded through hybrid sp^2 orbitals.

5. Applications:

- Actively used in cutting-edge technologies.
- Expected to play a vital role in future high-tech applications.

These nanocarbons exhibit exceptional properties, making them versatile and promising materials in various scientific and technological advancements.

NANOMATERIALS- FULLERINES...

1. **Discovery:** One type of fullerene, was discovered in 1985. Consists of a hollow spherical cluster of 60 carbon atoms, denoted as C₆₀.
2. **Structural Configuration:**
 - Carbon atoms form hexagonal and pentagonal geometrical configurations.
 - The C₆₀ molecule has 20 hexagons and 12 pentagons, resembling the symmetry of a soccer ball.
 - Material composed of C₆₀ molecules is known as buckminsterfullerene or buckyball.
3. **Solid State Arrangement:**
 - In the solid state, C₆₀ units form a crystalline structure in a face-centred cubic array.
 - This material is called fullerite.
4. **Endohedral and Exohedral Fullerenes:**
 - **Endohedral Fullerenes:** Atoms or groups of atoms encapsulated within the carbon cage.
 - **Exohedral Fullerenes:** Atoms, ions, or clusters of atoms attached to the outside of the fullerene shell.
5. **Applications:**
 - **Antioxidants:** Used in personal care products.
 - **Biopharmaceuticals:** Unique chemical and biological characteristics.
 - **Catalysts:** Applied in various chemical processes.
 - **Organic Solar Cells:** Utilized in renewable energy.
 - **Long-life Batteries:** Potential for enhanced battery performance.
 - **High-temperature Superconductors:** Advancements in superconductivity.
 - **Molecular Magnets:** Development of advanced magnetic materials.

NANOMATERIALS

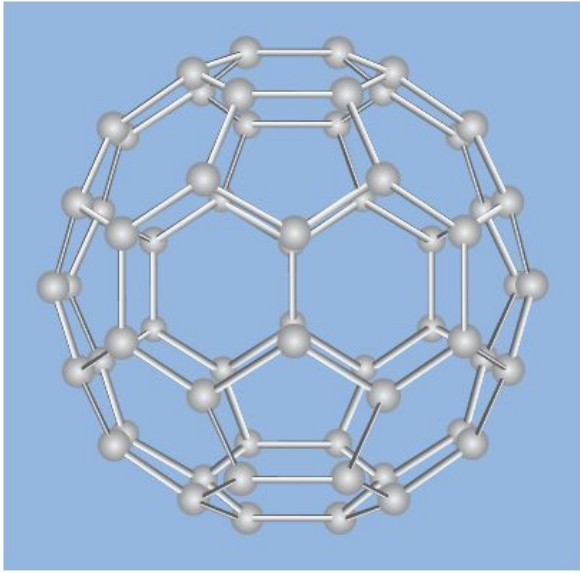


Figure 13.9 The structure of a C₆₀ fullerene molecule (schematic).

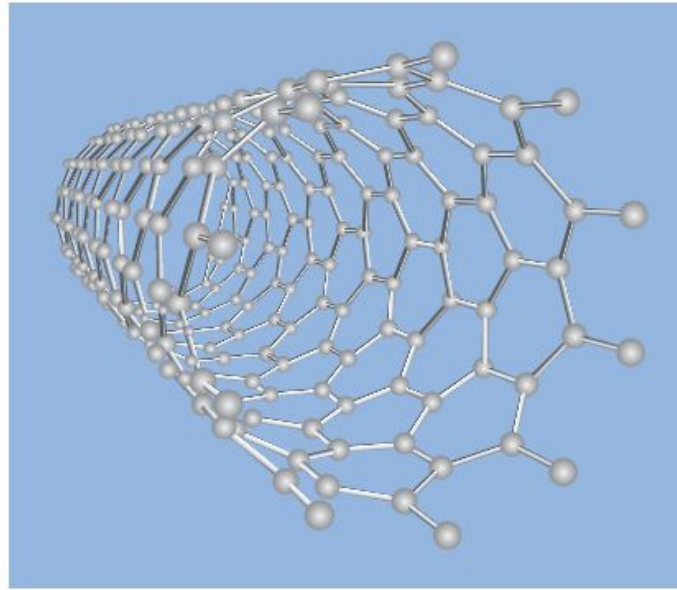


Figure 13.10 The structure of a single-walled carbon nanotube (schematic).

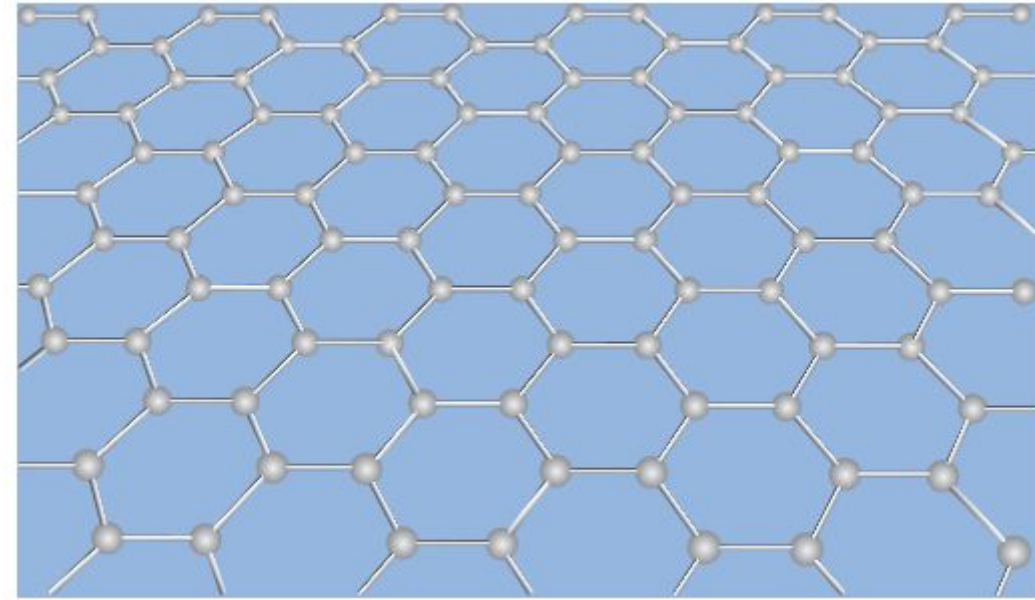


Figure 13.11 The structure of a graphene layer (schematic).

NANOMATERIALS- CARBON NANOTUBE

1. Structure:

- Single-walled carbon nanotube (SWCNT) consists of a single graphene sheet rolled into a tube.
- Multiple-walled carbon nanotubes (MWCNTs) have concentric cylinders.

2. Dimensions:

- Each nanotube is a single molecule composed of millions of atoms.
- Length is much greater (thousands of times) than its diameter.

3. Mechanical Properties:

- Extremely strong and stiff, relatively ductile.
- Tensile strengths range between 13 and 53 GPa (orders of magnitude greater than carbon fibres).
- Elastic modulus values around 1 TPa, with fracture strains between 5% and 20%.
- Relatively low densities.

NANOMATERIALS- CARBON NANOTUBE

4. Applications:

I. Structural Applications:

- Potential use in structural applications due to high strength.
- Current applications are limited to bulk nanotubes in nanocomposites to enhance mechanical, thermal, and electrical properties.

II. Electrical Characteristics:

- Metal or semiconductor behaviour based on hexagonal unit orientation.
- Potential use in wiring for circuits (metal state) or transistors and diodes (semiconductor state).
- Excellent electric field emitters for flat-screen displays.

III. Other Potential Applications:

- More efficient solar cells.
- Better capacitors as battery replacements.
- Heat removal applications.
- Cancer treatments (target and destroy cancer cells).
- Biomaterial applications (e.g., artificial skin, monitoring engineered tissues).
- Body armor.
- Municipal water-treatment plants (more efficient removal of pollutants and contaminants).

NANOMATERIALS- GRAPHENE...

1. Structure:

- Single atomic layer of graphite.
- Composed of hexagonally sp² bonded carbon atoms.

2. Production:

- Initially produced by peeling apart layers of graphite using plastic adhesive tape.
- Other cost-effective processes developed for high-quality graphene production.

3. Exceptional Characteristics:

- Perfect order with no atomic defects.
- Extremely pure with only carbon atoms.
- Unbonded electrons move much faster than in ordinary metals and semiconductors.

4. Properties:

- **Strength:** Strongest known material (~130 GPa).
- **Thermal Conductivity:** Best thermal conductor (~5000 W/m ·K).
- **Electrical Conductivity:** Lowest electrical resistivity (10⁻⁸ Ω·m), best electrical conductor.
- **Transparency:** Transparent.
- **Chemical Inertness:** Chemically inert.
- **Modulus of Elasticity:** Comparable to other nanocarbons (~1 TPa).

NANOMATERIALS- GRAPHENE

4. Potential Applications:

I. Electronics:

- Touch screens, conductive ink for electronic printing, transparent conductors, transistors, and heat sinks.

II. Energy:

- Polymer solar cells, catalysts in fuel cells, battery electrodes, supercapacitors.

III. Medicine/Biotechnology:

- Artificial muscle, enzyme and DNA biosensors, photoimaging.

IV. Aeronautics:

- Chemical sensors (for explosives), and nanocomposites for aircraft structural components.

5. Technological Potential:

- Enormous potential to revolutionize industries, including electronics, energy, transportation, medicine/biotechnology, and aeronautics.

7. Challenges:

- Economical and reliable methods for mass production are needed before the full realization of its potential.

BIOMATERIALS...

1. **Introduction:** Biomaterials are substances engineered to interact with biological systems, tissues, or organisms, often for medical purposes.
2. **Types of Biomaterials:**
 - I. **Metallic Biomaterials:**
 - i. Examples: Titanium, stainless steel.
 - ii. Applications: Orthopaedic implants, dental devices.
 - II. **Polymeric Biomaterials:**
 - i. Examples: Polyethylene, silicone.
 - ii. Applications: Artificial organs, and drug delivery systems.
 - III. **Ceramic Biomaterials:**
 - i. Examples: Hydroxyapatite, alumina.
 - ii. Applications: Dental implants, bone grafts.
 - IV. **Composite Biomaterials:**
 - i. Combinations of metals, polymers, and ceramics.
 - ii. Applications: Joint replacements, tissue engineering scaffolds.

BIOMATERIALS

4. Properties of Biomaterials:

- I. **Biocompatibility:** Ability to coexist with living tissues without causing adverse reactions.
- II. **Mechanical Properties:** Tailored to match the mechanical characteristics of natural tissues.
- III. **Bioactivity:** Ability to stimulate specific cellular responses for tissue regeneration.
- IV. **Degradation Rate:** Controlled degradation for temporary implants.

5. Applications:

- I. **Medical Devices:** Implants, prosthetics, pacemakers.
- II. **Tissue Engineering:** Scaffold materials for growing new tissues.
- III. **Drug Delivery Systems:** Controlled release of pharmaceuticals.
- IV. **Dental Applications:** Fillings, crowns, implants.

6. Challenges and Future Trends:

- I. **Biocompatibility Enhancement:** Ongoing research to improve compatibility.
- II. **Nanotechnology Integration:** Nanoscale biomaterials for targeted therapies.
- III. **3D Printing Technology:** Precision in fabricating complex biomaterial structures.

OPTICAL FIBERS...

1. Introduction:

Optical fibers are thin, flexible, transparent strands of glass or plastic used for transmitting light signals over long distances.

2. Structure of Optical Fibers:

- I. **Core:** Innermost part through which light travels. Composed of high-refractive-index glass or plastic.
- II. **Cladding:** Surrounds the core, designed with a lower refractive index. Facilitates light reflection within the core.
- III. **Buffer Coating:** Outermost layer for protection. Enhances durability and flexibility.

3. Principles of Light Transmission:

- I. **Total Internal Reflection:** Light reflects off the core-cladding interface. Ensures minimal signal loss during transmission.
- II. **Modulation:** Encoding data into light signals by varying intensity or frequency. Common modulation techniques: amplitude modulation, and frequency modulation.

4. Types of Optical Fibers:

- I. **Single-Mode Fibers:** Small core diameter, allows only one mode of light. Ideal for long-distance communication.
- II. **Multi-Mode Fibers:** Larger core diameter, permits multiple modes of light. Suitable for shorter-distance communication.
- III. **Plastic Optical Fibers (POF):** Made of plastic materials. Used for short-distance data transmission.

OPTICAL FIBERS

5. Applications:

- I. **Telecommunications:** Backbone networks, internet connections.
- II. **Data Transmission:** Local area networks (LANs), high-speed internet.
- III. **Medical Imaging:** Endoscopy, imaging devices.
- IV. **Sensors:** Temperature, pressure, and strain sensors.
- V. **Broadcasting:** Cable television, satellite communication.

6. Advantages of Optical Fibers:

- I. **High Bandwidth:** Supports large data transmission capacities.
- II. **Low Signal Loss:** Minimal attenuation over long distances.
- III. **Immunity to Electromagnetic Interference:** Unaffected by electrical noise.
- IV. **Secure Communication:** Difficult to tap or intercept.

7. Future Trends:

- I. **Fiber Optic Internet Expansion:** Wider deployment for faster internet connections.
- II. **5G Networks:** Integral for high-speed, low-latency communication.
- III. **Fiber Optic Sensing Technologies:** Advancements in applications like structural health monitoring